

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall ingest household monthly electricity (kWh) and transport (km) logs to compute total monthly carbon equivalent (CO2e) emissions.	High	Pass: Calculated emissions match EPA conversion factor formulas. Fail: Zero or negative emission values generated.	Core function of the tool; without this no CO2e figure exists to track.	
FR-002	Functional	The system shall allow the Resident to set a monthly or annual carbon-reduction target.	High	Pass: Target value is saved and correctly retrieved for the household on next login. Fail: Target is not persisted or is mixed up across households.	Enables the 'target tracking' half of the product, not just raw logging.	
FR-003	Functional	The system shall calculate and display the Resident's percentage progress toward their active reduction target after each new monthly log entry.	High	Pass: Progress percentage updates within the same session immediately after a log is submitted. Fail: Progress is stale, missing, or mathematically inconsistent with logged data.	Gives residents continuous feedback, which drives sustained engagement with the tool.	
FR-004	Functional	The system shall notify the Resident when a progress milestone (25%, 50%, 75%, 100% of target) is reached.	Medium	Pass: Exactly one notification fires the first time each milestone threshold is crossed. Fail: Duplicate notifications, or a milestone is skipped.	Motivational reinforcement; also used to demonstrate the <<extend>> relationship in the use-case model.	
FR-005	Functional	The system shall allow a Sustainability Coach to view a Resident's household consumption data and submit written recommendations linked to that household.	Medium	Pass: Recommendation is timestamped, stored, and visible to the correct Resident only. Fail: Recommendation is not linked to the right household or is visible to other households.	Adds the human-advisory layer that differentiates this from a plain calculator app.	
NFR-001	Nonfunc.	The annual carbon-reduction milestone chart must render interactive historical comparison graphs in under 200 ms, meeting security standards under simulated peak load.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Render time exceeds 200 ms or a security check fails under load.	Given requirement in the problem brief; slow or insecure charts undermine the analytics positioning.	
NFR-002	Nonfunc.	All household consumption data shall be encrypted at rest and in transit and accessible only to the owning Resident and their assigned Sustainability Coach.	High	Pass: Penetration test finds no unauthorized cross-household data access; data is encrypted with AES-256 or equivalent. Fail: Any household's data is readable by another household or an unassigned coach.	Utility and commute logs can reveal home-occupancy patterns, a genuine privacy risk worth an explicit NFR.	